

Solar Sponge VPP

RESOLVING SOLAR CURTAILMENT

A software-based flexible energy solution to
power New York through solar

Coco Guo, James Zhang, Pashka Durgaj, Shravani Survase





THE PROBLEM



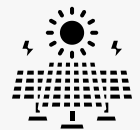
Old architecture makes clean energy tech more difficult to implement



Curtailment when solar power generation exceeds demand



Data centers require an abundance of energy to maintain cooling

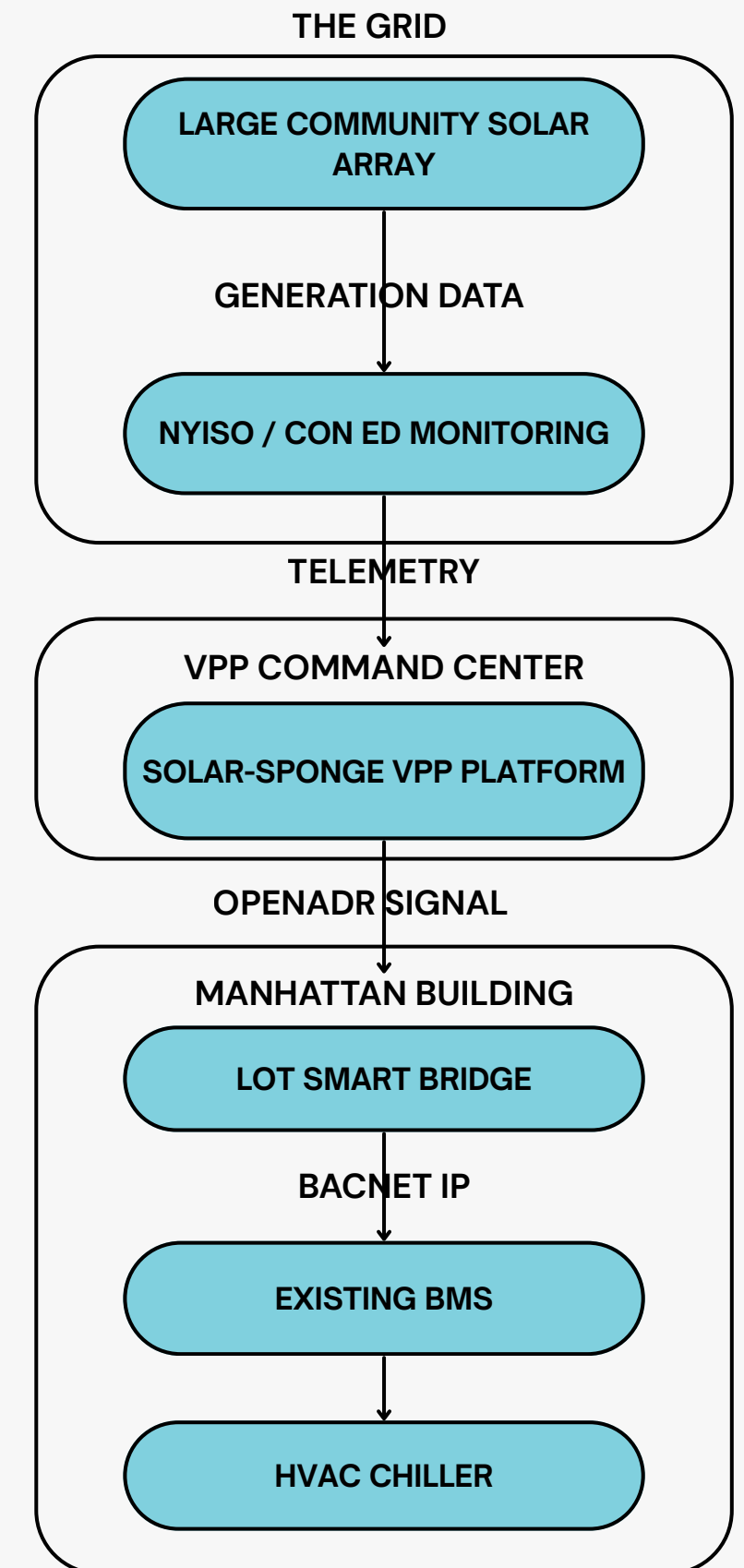


OUR SOLUTION

A "Solar Sponge" Virtual Power Plant

Implement a Virtual Power Plant (VPP) model that redistributes excess solar power to large buildings, particularly data centers, for thermal management.

Forego the need for expensive renovation and construction by installing non-invasive, cost effective IoT sensors to control HVAC via software.





PARTNERS & CLIENTS



Partner:

Con Edison

Smart meter infrastructure

Grid data & signals

Clients:

Data Centers

Provide a flexible load

Cooling systems absorb excess energy

Solar Power Generator

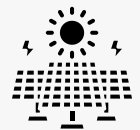
Share extra benefits

Reduce congestion & avoid grid upgrades

Building Owners

Lower energy costs

Reduce carbon penalties (LL97)



BUSINESS MODEL

Shore-Sponge buys solar credits from developers cheaply, resells them to data centres at a discount to grid rate, and keeps the spread – while layering on NYISO capacity payments and a SaaS dashboard fee.

Three stacked revenue lines from a single customer relationship. Data centres win on all three simultaneously.



Revenue per MW = 113K

- **CDG Spread**
- **SaaS (per MW basis)**



Cost per MW = \$71K

- **Solar developer pmt**
- **NYISO cuts**
- **Allocated engineering**



Gross margin per MW = \$42K (37%)



WHY THIS MATTERS

10,115

tonnes CO2 avoided per year = removing 2200 cars from NYC road for a year

40%

daytime carbon reduction for enrolled data centres

\$2.7mn

in LL97 fines avoided across portfolios per year



Solar Sponge VPP

THANK YOU! FOR LISTENING

Thank you for exploring our vision. Contact us today to discover how solar energy can transform your world.